GridSight

PAKISTAN POWER MARKET BRIEF

September 2026

System marginal price, national grid (ISMO)
Generated 21 Oct 2026 02:54 PKT
Data through 30 Sep 2026

Average price	Hours at Rs 50+	Night vs midday	Fuel-shock monitor
Rs 32.4	**227**	**Rs 40 / 16**	RED
+1% vs prior month · +98% y/y	32% of the month · prior month 217	00-06h vs 10-16h averages	Brent $117 (Sep), +37% in 3 months

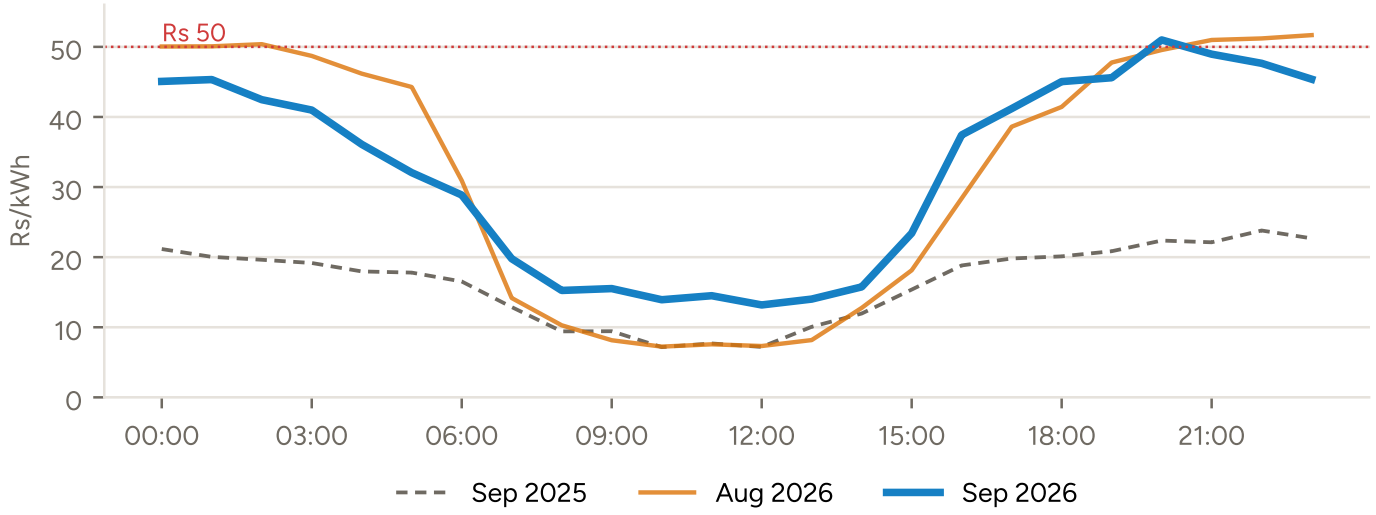
September 2026 averaged Rs 32.4/kWh, up 1% on the month before and up 98% on September 2025.

227 of its 720 hours (32%) were at or above Rs 50, with a high of Rs 93.9 at 21:00 on 08 Sep.

The day split in two: midday hours (10:00-16:00) averaged Rs 15.8 while the night (00:00-06:00) averaged Rs 40.3. Solar output pushes cheap plants onto the margin at midday; after sunset the system falls back on expensive thermal plants.

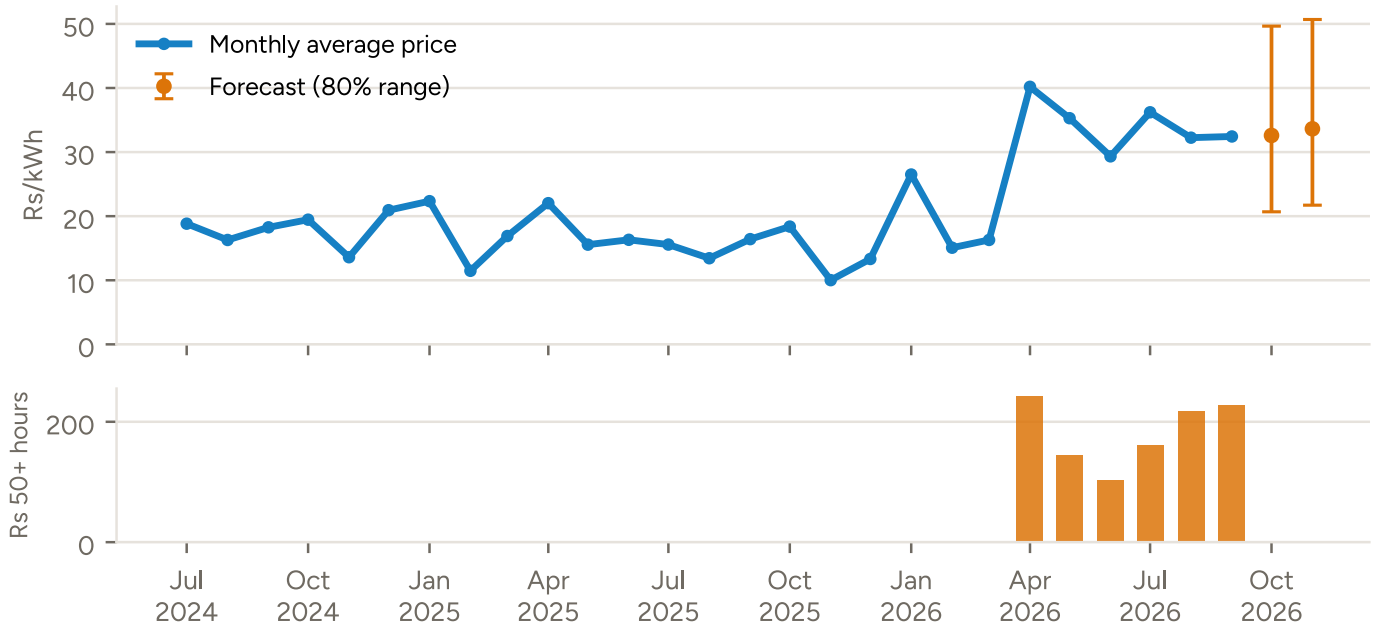
Fuel-shock monitor is RED: Brent averaged $117/bbl in September, +37% over three months.

The average day



Average price by hour of day. Hours are hour-beginning, Pakistan Standard Time.

Prices month by month, and the outlook



Top: monthly average price. Bottom: hours at Rs 50 or more. Forecast points are the next-month model's median and 80% range.

FUEL COST BY TECHNOLOGY

FUEL	FUEL COST, RS/KWH	PRIOR MONTH	TWH
Furnace oil (RFO)	52.2	42.2	0.10
Imported LNG (RLNG)	35.4	30.6	1.46
Imported coal	16.7	17.9	1.70
Thar coal	11.5	11.2	1.36
Local gas	13.7	13.6	0.87
Nuclear	2.9	2.7	1.80

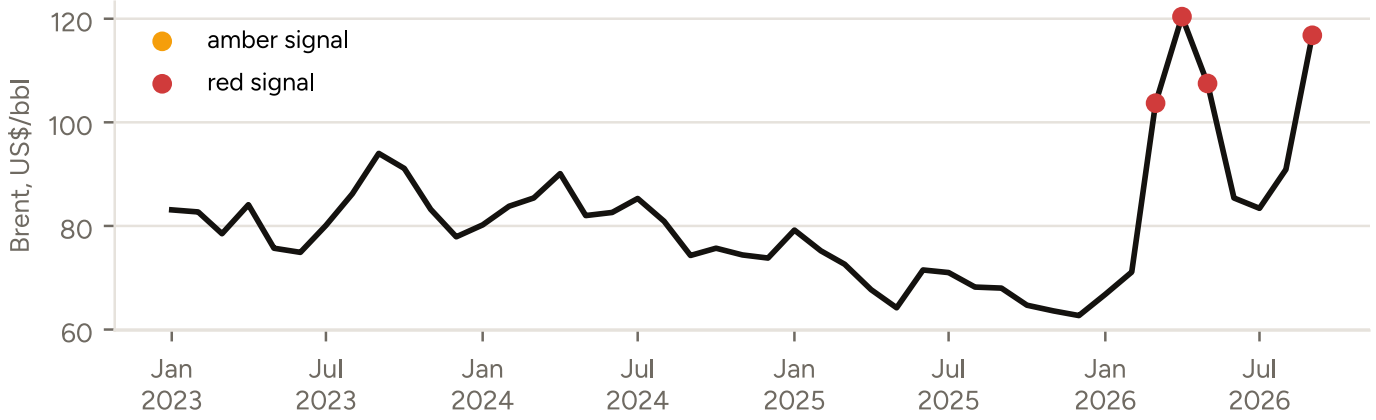
CPPA-G energy purchase data for June 2026 (published with a lag). Fuel charges divided by energy bought; the most expensive fuel that runs sets the evening price.

FUEL-SHOCK MONITOR: RED

Brent averaged **$116.8/bbl** in September 2026: +28% on the month, +37% over three.

LNG (Japan), US$/mmbtu	14.2	+21% (3m)
Coal (Australia), US$/t	147.1	+6% (3m)
Gas (Europe), US$/mmbtu	25.4	+68% (3m)

Why oil: the Rs 50+ hours are furnace-oil plants setting the price. Brent rose 46% in March 2026, the month before those hours began.



Red: Brent at or above $90 after rising 30%+ in three months. Amber: a rise of 30%+ in three months or 20%+ in one, from $75+. Since 2018 it has fired in: Mar 2022 (red), Mar 2026 (red), Apr 2026 (red), May 2026 (red), Sep 2026 (red). With one regime

change in the price history, this is a monitor, not a validated predictor.

WHICH PLANT SET THE PRICE

Coal plants set the price in 25% of hours; in the evening (18:00 to midnight) it was mostly imported lng (33%). The plant that set it most often was KEL (K-Electric), on furnace oil, in 76 hours.

FUEL	ALL HOURS	EVENING
Coal	25%	7%
Imported LNG	22%	33%
Furnace oil	20%	29%
Local gas	8%	1%
Diesel	6%	13%
Low-sulphur furnace oil	2%	5%
Furnace oil/gas mix	1%	0%
Between two fuels	17%	12%

PLANT	HOURS	RS/KWH
KEL (K-Electric)	76	54.10
Jamshoro (Coal)	56	15.02
747 MW GUDDU (OC)	40	13.87
Thar Coal Block-I	38	5.39
Sahiwal Power	29	20.52

Each hour's official price matched to ISMO's merit order in force (plants ranked by running cost). Exact plant known for 53% of hours, fuel for 83%; the rest lie between plants on different fuels. Evening = 18:00 to midnight.

Outlook

MONTH	AVERAGE, RS/KWH	80% RANGE	MIDDAY (10-16H)	EVENING (18-24H)	USES PRICES TO
October 2026	32.6	20.7 - 49.7	22.0	40.8	end of August 2026
November 2026	33.6	21.7 - 50.7	18.7	45.8	end of September 2026

Next-month model: the same-hour average of the last three published months, with a range from its own past errors. ISMO publishes a month's prices 2 to 5 weeks after it ends, so a forecast for month T can only use prices to the end of T-2. The range is the average of hourly ranges; single hours spread wider.

> **Day ahead, 01 October 2026:** median Rs 21.5/kWh across the day; 0 hours expected at Rs 50+; 0 possible, where the upper band crosses Rs 50. For a participant with daily price data; the public file arrives monthly.

HOW THE FORECASTS HAVE DONE

MODEL	HORIZON	ERROR, SEP 2026	ERROR, 2026 TEST	80% RANGE HELD
Our live model (blend)	day ahead	Rs 8.74	Rs 7.80	71%
Same hour yesterday	day ahead	Rs 9.41	Rs 8.10	69%
Our model (3-month profile)	next month	Rs 15.47	Rs 13.37	55%
Same hour, last known month	next month	Rs 15.21	Rs 15.33	50%
ISMO's CTBCM projection	next month	Rs 18.36	Rs 15.00	

Error: mean absolute difference from the published price, Rs/kWh per hour. Walk-forward: every forecast used only data available at the time. The 2026 test (January onward) was never used to choose models, except the blend, chosen after it; its forward record starts September 2026. ISMO's projection, built on September 2025 inputs, is compared at the monthly horizon.

For bulk consumers weighing open access

At September's prices, a 5 MW B-3 factory running around the clock would have paid **Rs 48.7/kWh** buying at the hourly market price through the wheeling auction, against **Rs 30.6/kWh** on its DISCO tariff: Rs 18.1/kWh more. Energy alone, grossed up for network losses, came to Rs 36.3/kWh.

> The highest fixed contract price at which open access still breaks even for this factory: **Rs 18.9/kWh** at the generator through the auction, **Rs 7.3/kWh** outside it (where the Rs 12.94 stranded-cost charge applies).

Grid charges per NEPRA's uniform Use of System Charges decision (7 Sep 2026); losses, capacity and ancillary costs as in ISMO's CTBCM model; regulated tariff per NEPRA decision TRF-100 of 11 Feb 2026. Excludes sales tax and duty, which apply to both routes.

Data quality

The latest automated check of the public source data found 3 critical issues and 22 warnings; the critical ones:

ismo monthly profiles	Published as Jul 2024 but the whole workbook is Jun 2024's (identical to the file published for that earlier month); file a56484f3, posted in 5 section(s): DISCO's Consumption, Fuel Mix, Hourly Generation Profile, Marginal Price vs Demand, System at Glance. Anyone using ISMO's site for Jul 2024 gets the wrong month.
ismo monthly profiles	Published as Jan 2025 but the whole workbook is Mar 2025's (identical file also published as that month, and it fits that month's hourly prices); file f2cf7e0b, posted in 1 section(s): Marginal Price vs Demand. Anyone using ISMO's site for Jan 2025 gets the wrong month.
ismo monthly profiles	Published as Apr 2025 but the whole workbook is Mar 2025's (identical file also published as that month, and it fits that month's hourly prices); file f2cf7e0b, posted in 4 section(s): DISCO's Consumption, Fuel Mix, Hourly Generation Profile, System at Glance. Anyone using ISMO's site for Apr 2025 gets the wrong month.

Sources

FILE	PUBLISHED	SHA-256
Hourly Marginal Price July 2024 to Sep 2026	15 Oct 2026	`e41a569a9b51b498`
CS Bilateral Transaction Model	22 Sep 2026	`7c2887b180765187`
CMO Historical Data Monthly		`ea1c350827878ea3`
Decision of Authority - Use of System Charges	15 Sep 2026	`ef0a730b660e1cc1`

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